

Identity-Preserving Adaptive Surveillance Restoration & Recognition Pipeline

Final Year Project Proposal

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BS Artificial Intelligence



Department of AI

Fast School of Computing

Fast National University, Karachi Campus

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Project Registration

Program: BS Artificial Intelligence (all members)

Supervisor: Dr. Kamran Ali Co-supervisor: None

Project ID (for office use)						
Type of project	☒ Traditional ☐ Industrial ☐ Continuing					
Nature of project	☐ Development ☒ Research & Development ☐ Research					
Sustainable Development Goals(SDGs)	☐ Good Health and Well-Being ☒ Quality Education ☒ Industry, Innovation, and Infrastructure ☐ Gender Equality ☐ Decent Work and Economic Growth ☐ Climate Action					
Area of specialization	☒ Artificial Intelligence (AI) ☒ Data Science and Analytics ☐ Internet of Things (IoT) ☐ Blockchain ☐ Mobile App Development ☐ Web Development ☐ Cybersecurity ☐ Game Development ☐ Natural Language Processing (NLP) ☐ Other					
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Declaration: FYP group members have cleared all prerequisite courses For FYP-I as per their degree requirements.						

1. Project Abstract

Surveillance cameras are widely used for public safety, but low resolution, motion blur, compression artifacts, and poor lighting often degrade captured faces and reduce recognition reliability. This project proposes an identity-preserving adaptive surveillance restoration and recognition pipeline that focuses on improving recognition utility without introducing identity-altering hallucinations. The core research gap addressed is the limited prior work on combining three elements in a single framework: conditional quality-gated restoration (skip or run enhancement only when needed), full stage-wise baseline-versus-frontier ablation across restoration and recognition modules, and evaluation on surveillance-native face benchmarks rather than only synthetic degradation datasets.

The proposed Approach B pipeline begins with ViT-based frame quality assessment, then conditionally applies diffusion-based restoration modules: DreamSR or InvSR for super-resolution,

ID-CDM for zero-shot deblurring, and π -Diff for low-light enhancement. Restored and non-restored streams are passed through face detection (RetinaFace and SFE-DETR), alignment, and recognition ablations (ArcFace, AdaFace, MagFace, CurricularFace, and PartialFC), followed by FAISS/HNSW retrieval and BoT-SORT temporal tracking.

Evaluation is explicitly identity-aware: image metrics (PSNR, SSIM, LPIPS) are reported alongside downstream recognition metrics (Rank-1 and TAR@FAR) with pre/post-restoration comparisons. Datasets include QMUL-SurvFace and QMUL-TinyFace as headline surveillance-native benchmarks, supported by WIDER FACE, IJB-B/IJB-C, RealBlur, HIDE, LOL/LOL-v2, and SICE for module-level validation. Expected outcomes include a reproducible adaptive pipeline, rigorous ablation evidence, and practical guidance on when modern diffusion restoration improves recognition and when it harms identity fidelity.

2. Introduction

This project lies at the intersection of computer vision, image restoration, and face recognition for surveillance analytics. In operational CCTV settings, recognition systems must process uncontrolled imagery with severe degradations, making identity retrieval difficult even with modern recognizers.

The domain is surveillance face analytics under real-world quality constraints. It is important because inaccurate recognition in public-safety workflows can reduce trust, increase manual review burden, and limit deployment value. Current pipelines often optimize restoration quality visually but do not consistently test whether restored faces preserve identity information required by recognizers, see Fig 3. AI-driven methods are relevant because recent diffusion and quality-adaptive recognition models can recover useful detail and handle mixed-quality inputs, but they also risk introducing artifacts that alter identity cues.

Fig 1. shows relevant literature review and Fig 4. shows datasets. Note that the final pipeline will need only test data since each module is independently trained.

3. Problem Statement

The research problem is to determine whether diffusion-based restoration can improve surveillance face recognition on real low-quality CCTV imagery while preserving identity-discriminative information. The methodological gap is not the absence of software, but the absence of a unified adaptive pipeline that conditionally applies restoration, performs end-to-end stage ablations, and evaluates restoration impact using identity-aware metrics on surveillance-native datasets.

4. Motivation

The motivation is to bridge the gap between visually improved faces and truly recognizable faces in surveillance scenarios, where degraded inputs are the norm and identity errors carry operational risk.

The project is motivated by the need to:

- reduce manual filtering and repeated review of unusable face captures.
- improve automated triage by routing only low-quality frames to restoration.

- • automate restoration-detection-recognition-tracking stages in one reproducible workflow.
- • produce evidence-based guidance for choosing restoration strength under identity constraints.
- • improve recognition robustness on surveillance-native benchmarks.
- • support centralized, searchable embeddings for identity retrieval.

a. Aims and Objectives

Goal: design and validate an adaptive, identity-preserving surveillance restoration and recognition pipeline using finalized Approach B, probable system requirements shown in Fig 2.

Objectives:

- 1) Build a conditional quality-gated restoration flow using DreamSR/InvSR, ID-CDM, and π -Diff.
- 2) Compare RetinaFace and SFE-DETR for degraded-face detection.
- 3) Run recognition ablations across ArcFace, AdaFace, MagFace, CurricularFace, and PartialFC.
- 4) Quantify restoration impact with both image-quality and identity-aware metrics.
- 5) Benchmark on QMUL-SurvFace and QMUL-TinyFace with supporting datasets for stage-wise analysis.

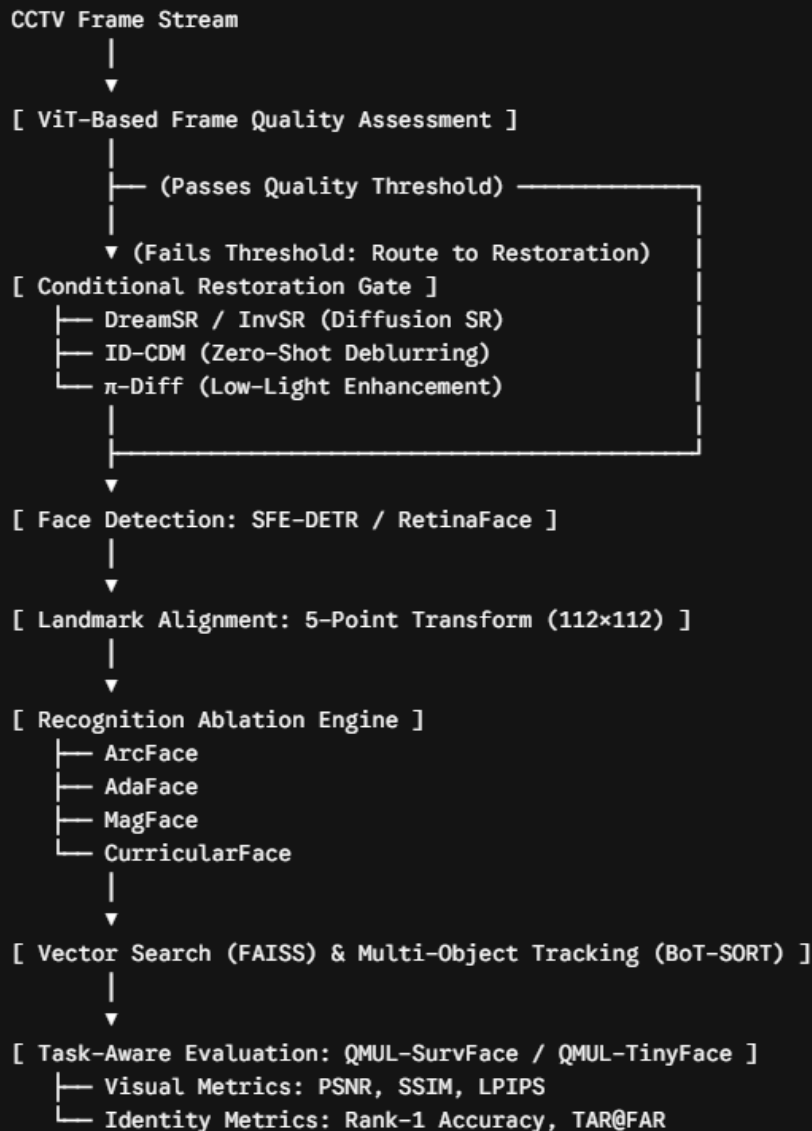
b. Scope of the Project

In scope: quality assessment, conditional restoration, face detection/alignment, recognition ablations, embedding search, and temporal tracking with reproducible evaluation protocols. The main empirical focus is surveillance-native performance on QMUL-SurvFace and QMUL-TinyFace, supported by module-level tests on RealBlur, HIDE, LOL/LOL-v2, SICE, WIDER FACE, and IJB-B/IJB-C. Out of scope: non-face traffic analytics branch and deployment on event-camera hardware.

5. Proposed Methodology

Methodology follows a stage-by-stage pipeline with adaptive control. Step 1: estimate frame quality using ViT-based quality assessment and thresholding. Step 2: apply restoration conditionally—super-resolution (DreamSR/InvSR), deblurring (ID-CDM), and low-light enhancement (π -Diff)—only when quality flags require correction. Step 3: detect faces using RetinaFace and SFE-DETR and align crops to a standard recognition input format. Step 4: extract embeddings with ArcFace, AdaFace, MagFace, CurricularFace, and PartialFC variants. Step 5: perform identity retrieval via FAISS/HNSW and maintain temporal identity continuity using BoT-SORT. Step 6: execute ablations across restoration/no-restoration and model combinations, then compare gains and failure modes.

Proposed Methodology



Functional requirements include modular stage execution, conditional restoration gates, multi-model ablation support, and identity retrieval/tracking outputs. Non-functional requirements include reproducibility, transparent metric reporting, manageable inference latency through selective restoration, and hardware-aware configuration for high-VRAM GPU environments.

6. System Architecture

The proposed system follows a modular pipeline architecture with interchangeable model components and experiment configuration control. Major components include data ingestion, quality assessment

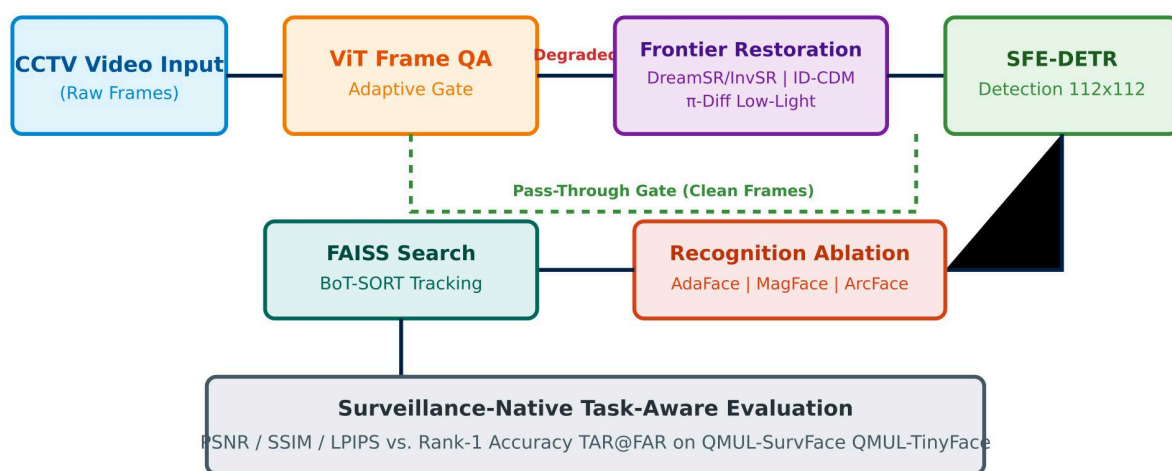
and gating, restoration engines, detection/alignment, recognition embeddings, vector search, temporal tracking, and evaluation/reporting modules.

Data and Experiment Layer – dataset loaders, preprocessing, run configuration, and logging.

Model Execution Layer – quality estimation, restoration, detection, alignment, recognition, retrieval, and tracking.

Evaluation Layer – PSNR/SSIM/LPIPS plus Rank-1 and TAR@FAR with pre/post-restoration comparisons.

Workflow: CCTV frames → quality assessment → conditional restoration → face detection/alignment → recognition embeddings → FAISS/HNSW matching → BoT-SORT identity tracking → identity-aware evaluation.



7. Individual Tasks

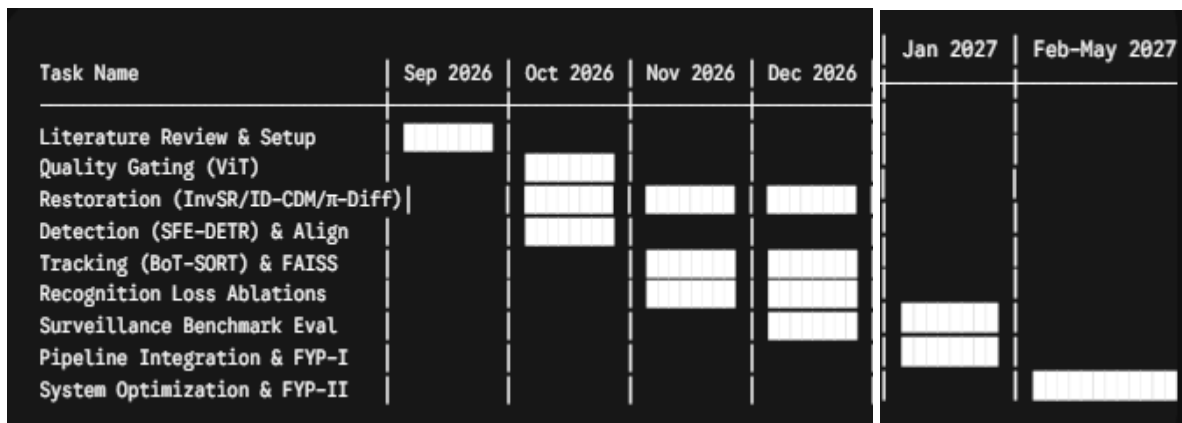
The group will divide responsibilities across restoration research, recognition/evaluation, and integration/deployment while sharing reproducibility and documentation duties.

Team Member	Activity	Tentative Date
Muhammad Abser Mansoor	Pipeline integration, quality gating design, and end-to-end experiment orchestration	Sep 2026 - Jan 2027
Raghib Rizwan Rabbani	Restoration module benchmarking (DreamSR/InvSR, ID-CDM, π -Diff) and optimization	Sep 2026 - Feb 2027
Muhammad Umar	Detection/recognition ablations, FAISS/HNSW	Sep 2026 - Feb 2027

	matching, and metric evaluation	
All Members	QMUL-SurvFace/TinyFace final evaluation, report writing, and presentation preparation	Feb 2027 - May 2027
All Members	Final revisions, documentation freeze, and submission	May 2027 - Jun 2027

8. Gantt Chart

Work will run in three phases: (1) literature replication and dataset preparation, (2) module integration and controlled ablations, and (3) final benchmarking, analysis, and report completion. Restoration and recognition experiments run partly in parallel, followed by joint end-to-end validation and thesis write-up.



9. Tools and Technologies

Primary stack: Python, PyTorch 2.3+, OpenCV, diffusers, torchvision, and NumPy. Models/libraries: DreamSR or InvSR, ID-CDM, π -Diff, RetinaFace, SFE-DETR, ArcFace/AdaFace/MagFace/CurricularFace/PartialFC implementations, FAISS or HNSW indexing, and BoT-SORT tracking. Experiment support: Jupyter, Weights & Biases or TensorBoard for logging, and Git/GitHub for reproducibility. Compute target aligns with Approach B requirements (high-VRAM NVIDIA GPUs, CUDA 12.x).

10. Expected Deliverables

- 1) Completed adaptive surveillance restoration-recognition pipeline (Approach B).
- 2) Stage-wise and end-to-end ablation results across restoration and recognition models.

- 3) Identity-aware evaluation report on QMUL-SurvFace and QMUL-TinyFace with supporting benchmarks.
- 4) Reproducible experiment configs, trained checkpoints (where permissible), and implementation documentation.
- 5) Final FYP report and presentation material summarizing findings, limitations, and future extensions.

11. References

- [1] Y. Dong et al., “DreamSR: Towards Ultra-High-Resolution Image Super-Resolution via a Receptive-Field Enhanced Diffusion Transformer,” CVPR, 2026.
- [2] J. Wang, B. Chen, and H. Dai, “Zero-Shot Realistic Image Deblurring with Consistency Model (ID-CDM),” Complex & Intelligent Systems, 2025.
- [3] X. Wang et al., “ π -Diff: Physically-Inspired Low-Light Image Enhancement with Structure-Preserving Diffusion Priors,” CVPR Workshop (LoViF), 2026.
- [4] J. Yang, J. Jiang, and T. Song, “SFE-DETR: An Enhanced Transformer-Based Face Detector for Small Target Faces in Open Complex Scenes,” Sensors, 2025/2026.
- [5] M. Kim et al., “AdaFace: Quality Adaptive Margin for Face Recognition,” CVPR, 2022.
- [6] X. Wang et al., “QMUL-SurvFace: Face Recognition Challenge in Unconstrained Surveillance Environments,” 2019.
- [7] P. Zhu et al., “QMUL-TinyFace: A Large-Scale Native Low-Resolution Face Recognition Benchmark,” 2021.

12. Figures and Tables

Fig 1 Stage-by-Stage Literature Table

Stage	Task	Recommended Model(s)	Key 2025–2026 Paper(s) (verified)	Why Chosen	Advantages	Disadvantages	Relevant Dataset(s)	Implementation Difficulty
1	Frame quality assessment	CNN/ViT-based blur, noise, exposure, and compression estimators	No single dominant 2025–2026 "surveillance QA" paper; this stage is typically assembled from classical + learned no-reference IQA (e.g., Laplacian variance for blur, NIQE/BRISQUE-style learned successors)	Prevents unnecessary restoration, which is the single biggest source of identity distortion in a naive pipeline	Cheap, fast, gates expensive stages	No single agreed-upon SOTA model; must be assembled/validated by the student	Custom-labelled subset of your CCTV data; RealBlur/GoPro for blur-score calibration	Medium — mostly engineering, some threshold-tuning
2A	Super-resolution (SR) —	Real-ESRGAN	Wang et al., <i>Real-ESRGAN: Training</i>	Mature, fast (GAN,	Fast inference, small	Can hallucinate texture;	RealSR, DIV2K+d	Low

Stage	Task	Recommended Model(s)	Key 2025–2026 Paper(s) (verified)	Why Chosen	Advantages	Disadvantages	Relevant Dataset(s)	Implementation Difficulty
	production baseline		<i>Real-World Blind Super-Resolution with Pure Synthetic Data</i> , ICCVW 2021	single forward pass), open-source, widely used on CCTV-style low-res footage	VRAM footprint, easy to fine-tune	weaker fidelity than diffusion SR on faces	pipeline, your own CCTV crops	
2A	Super-resolution — frontier comparator	DreamSR (diffusion transformer, receptive-field enhanced)	Dong et al., DreamSR: Towards Ultra-High-Resolution Image Super-Resolution via a Receptive-Field Enhanced Diffusion Transformer , CVPR 2026 (arXiv:2605.15682)	Confirmed CVPR 2026 paper; addresses the specific failure mode of patch-wise diffusion SR (local over-generation) that is most damaging to small, cropped face regions	Sharper, more faithful local texture than earlier diffusion SR; explicitly targets the "patch hallucination" problem	Multi-step diffusion inference, heavier VRAM, slower	Its evaluation benchmarks + your CCTV faces	High
2A	Super-resolution — efficient one-step alternative	FluxSR / TADSR / InvSR	FluxSR: Li et al., <i>One Diffusion Step to Real-World Super-Resolution via Flow Trajectory Distillation</i> , ICML 2025; TADSR: <i>Time-Aware One-Step Diffusion Network for Real-World Image</i>	All three confirmed as genuine 2025 one-step/few-step diffusion distillation papers — the	1–5 sampling steps instead of 20–50; near-diffusion quality at near-GAN speed	Younger codebases, less battle-tested on CCTV-specific degradations	Same as above; InvSR reports on RealSet80/ImageNet-Test	Medium-High

Stage	Task	Recommended Model(s)	Key 2025–2026 Paper(s) (verified)	Why Chosen	Advantages	Disadvantages	Relevant Dataset(s)	Implementation Difficulty
			<i>Super-Resolution</i> , arXiv:2508.16557 (2025); InvSR: Yue et al., <i>Arbitrary-steps Image Super-resolution via Diffusion Inversion</i> , CVPR 2025	realistic middle ground between Real-ESRGAN's speed and DreamSR's quality				
2B	Blur detection (gate)	Laplacian variance / FFT spectral energy / lightweight CNN classifier	No dedicated 2025–2026 paper — this is a well-established classical technique, correctly described as such	Sub-millisecond, no training required for the classical version	Trivial to implement, interpretable threshold	Classical variance methods are noise-sensitive; CNN variant needs labelled data	Any blur/sharp paired set (GoPro, RealBlur) for threshold calibration	Low
2C	Deblurring	ID-CDM (zero-shot consistency-distillation deblurring)	Wang, Chen & Dai, Zero-Shot Realistic Image Deblurring with Consistency Model (ID-CDM) , <i>Complex & Intelligent Systems</i> , 2025	Confirmed 2025 publication; explicitly zero-shot (no deblurring-specific training needed) and evaluated on a real-world blur set, which matters because synthetic-Gaussian-trained	Avoids paired-data training entirely; targets real (not synthetic) blur; faster than full multi-step diffusion via consistency distillation	Still diffusion-based → slower than a CNN deblurrer; identity-preservation on faces not specifically validated in the paper	HIDE (synthetic), RealBlur (real-world) — as used in the paper itself	High

Stage	Task	Recommended Model(s)	Key 2025–2026 Paper(s) (verified)	Why Chosen	Advantages	Disadvantages	Relevant Dataset(s)	Implementation Difficulty
				deblurrers generalize poorly to CCTV motion blur				
2C	Deblurring — efficiency benchmark context	Efficient/event-based deblurring challenges	AIM 2025: Feijoo et al., Efficient Real-World Deblurring using Single Images: AIM 2025 Challenge Report , ICCVW 2025 (arXiv:2510.12788) ; NTIRE 2026: The Second Challenge on Event-Based Image Deblurring at NTIRE 2026	Both confirmed as real, recent challenge-report papers. AIM 2025 caps models at <5M parameters / <200 GMACs — directly relevant if you need edge/real-time deployment. The NTIRE 2026 challenge needs event-camera input, which ordinary CCTV does not have, so it is context/b	Gives you a credible "efficient deblurring" citation and parameter/GMAC budget to benchmark against	Event-based NTIRE work is not applicable unless your camera has an event sensor — flag this in your literature review so an examiner doesn't think you deployed it	RSBlur (AIM 2025), HighREV (NTIRE event challenge)	Medium (AIM baseline), N/A for event-based unless you have event-camera hardware

Stage	Task	Recommended Model(s)	Key 2025–2026 Paper(s) (verified)	Why Chosen	Advantages	Disadvantages	Relevant Dataset(s)	Implementation Difficulty
				enchmarking material rather than a directly deployable stage				
2D	Low-light enhancement	π -Diff (recommended frontier) / Zero-DCE++ (baseline)	Wang et al., π-Diff: Physically-Inspired Low-Light Image Enhancement with Structure-Preserving Diffusion Priors , CVPR 2026 Workshop (LoViF); Li et al., <i>Zero-DCE++</i> , TPAMI 2021	π -Diff is a training-free diffusion framework designed specifically to avoid hallucinated content, color distortion, structural degradation, and over-enhancement in extremely dark scenes. It aligns closely with the identity-preservation goals of the overall surveillance pipeline.	π -Diff: strong structure preservation, no training required, diffusion-quality enhancement. Zero-DCE++: extremely fast and lightweight.	π -Diff is slower than Zero-DCE++; workshop publication rather than main-conference paper.	LOL, LOL-v2, SICE; NTIRE 2026 Efficient Low-Light Enhancement Challenge	Medium

Stage	Task	Recommended Model(s)	Key 2025–2026 Paper(s) (verified)	Why Chosen	Advantages	Disadvantages	Relevant Dataset(s)	Implementation Difficulty
				Zero-DC E++ remains the lightweight practical baseline.				
3	Face detection	RetinaFace (baseline) / SFE-DETR (frontier)	RetinaFace: Deng et al., CVPR 2020; Yang, Jiang & Song, SFE-DETR: An Enhanced Transformer-Based Face Detector for Small Target Faces in Open Complex Scenes , Sensors 2025/2026	SFE-DETR targets small, degraded, dense, and partially occluded faces in open complex scenes. These characteristics closely match CCTV surveillance environments and provide a modern transformer-based comparison against RetinaFace.	Better small-face performance, transformer architecture, reduced parameter count, stronger degraded-image robustness.	Sensors venue rather than CVPR/ICCV-tier conference; newer ecosystem and fewer implementations.	WIDER FACE	Medium
4	Face alignment	5-point / 68-point landmark-based similarity	Standard ArcFace-family preprocessing convention, not a separate paper	Well-defined, deterministic, and the de	Nearly free computationally; improves	Fails ungracefully on extreme	Same as detection stage	Low

Stage	Task	Recommended Model(s)	Key 2025–2026 Paper(s) (verified)	Why Chosen	Advantages	Disadvantages	Relevant Dataset(s)	Implementation Difficulty
		transform to 112×112		factor input contract for ArcFace/AdaFace-family recognize rs	recognition more than adding another restoration stage	pose/occlusion		
5	Face recognition (embedding)	ArcFace / AdaFace / MagFace / CurricularFace	ArcFace: CVPR 2019; AdaFace: CVPR 2022; MagFace: CVPR 2021; CurricularFace: CVPR 2020	Provides a complete comparison between classical margin-based recognition, quality-adaptive recognition, quality-aware representation learning, and curriculum-based hard-sample optimization.	Strong ablation study covering the major modern face-recognition paradigms.	Requires additional experimentation and evaluation time.	MS1MV3, Glint360K, LFW, CFP-FP, AgeDB, IJB-B, IJB-C, QMUL-SurvFace, QMUL-TinyFace	Medium-High
6	Identity tracking across frames	ByteTrack or BoT-SORT (associate face/person boxes over time)	ByteTrack: Zhang et al., ECCV 2022; BoT-SORT: Aharon et al., arXiv 2022	Both are the standard, still-dominant multi-object trackers layered	Reduces redundant recognition calls; gives persistent IDs across frames	Track fragmentation under long occlusion; not identity-aware by itself	MOT17, MOT20 (generic benchmark); your own CCTV footage for	Medium

Stage	Task	Recommended Model(s)	Key 2025–2026 Paper(s) (verified)	Why Chosen	Advantages	Disadvantages	Relevant Dataset(s)	Implementation Difficulty
				on top of a detector; not 2025–2026 but appropriately cited as established infrastructure		(needs the recognizer's embedding to actually confirm identity, not just track continuity)	qualitative evaluation	
7	Database matching	Cosine similarity / ANN search (FAISS, HNSW) over embeddings	Not paper-specific; standard vector-search engineering	Sub-linear search at scale	Simple, well-documented libraries	Threshold selection (FAR/FRR trade-off) needs careful evaluation, not just "similarity > 0.5"	Your enrolled-identity gallery + QMUL-SurvFace as an open-set stress test	Low-Medium
—	Non-face object detection /tracking (traffic)	YOLOv11 / RT-DETR + ByteTrack/ BoT-SORT	YOLOv11: Ultralytics, 2024 release (not a peer-reviewed paper — cite the Ultralytics documentation/Git Hub, not a nonexistent "YOLOv11 paper"); RT-DETR: Zhao et al., CVPR 2024	Independent branch from the face pipeline, as in your original design	Real-time, mature ecosystem	YOLOv11 has no formal peer-reviewed paper as of writing — this is a citation-accuracy issue you should flag explicitly to your supervisor	BDD100K, UA-DETRAC	Low

Fig 2 System requirements

Resource	Minimum	Recommended
GPU	NVIDIA RTX 4090 (24 GB VRAM)	A100 40GB / A6000 48GB (university HPC cluster access strongly recommended)
CPU	8-core	16-core
RAM	32 GB	64 GB
Storage	500 GB NVMe SSD	1–2 TB NVMe SSD (diffusion checkpoints alone can be 5–15 GB each)
CUDA/cuDNN	CUDA 12.x	CUDA 12.4+
Framework	PyTorch 2.3+, diffusers, xformers/flash-attention	Same, plus TensorRT or torch.compile for the recognition/detection stages to keep the pipeline usable
Expected throughput	1–4 FPS with full multi-step diffusion SR	5–10 FPS with one-step distilled variants (FluxSR/TADSR/InvSR at 1 step) on frames flagged as needing restoration
Cloud fallback	Single A100 40GB instance	Multi-GPU (2× A100) if you also want to run ablations in parallel

Fig 3 Classical CNN/GAN vs. Diffusion/Foundation-Model Comparison

Dimension	Classical CNN/GAN (Real-ESRGAN, RetinaFace, ArcFace, ByteTrack)	Diffusion/Foundation-Model (DreamSR, FluxSR/TADSR/InvSR, ID-CDM)
Inference speed	Milliseconds per frame, single forward pass	Multi-step: seconds per frame; one-step distilled variants: tens–hundreds of ms
VRAM	2–6 GB typical	8–24 GB depending on resolution and step count

Dimension	Classical CNN/GAN (Real-ESRGAN, RetinaFace, ArcFace, ByteTrack)	Diffusion/Foundation-Model (DreamSR, FluxSR/TADSR/InvSR, ID-CDM)
Texture/detail quality	Can look "plasticky" or over-smoothed; GANs can hallucinate implausible detail	Generally higher perceptual quality, better texture realism
Identity preservation	Predictable, but limited detail recovery on very low-res faces	Risk of <i>inventing</i> facial detail not present in the source — a serious concern for evidentiary/forensic use, must be evaluated with downstream recognition accuracy, not just PSNR/SSIM
Maturity / reproducibility	Very mature, well-documented, easy for an FYP timeline	Younger codebases (2025–2026), less tooling, more debugging overhead
Best FYP fit	Practical/reproducible pipeline (Approach A)	Frontier/publication-level pipeline (Approach B)

Fig 4 Dataset Table

Purpose	Dataset(s)	Notes
Super-resolution training/eval	RealSR, DIV2K+degradation, RSBlur (also usable for SR-adjacent eval)	Standard SR benchmarks; none are surveillance-native
Deblurring	GoPro (synthetic), HIDE (synthetic), RealBlur (real-world)	ID-CDM and most deblurring papers evaluate on this pair
Low-light enhancement	LOL, LOL-v2, SICE	Standard low-light benchmarks
Face detection	WIDER FACE	Standard, includes small/occluded faces relevant to CCTV
Face recognition — clean	LFW, CFP-FP, AgeDB, CALFW	High-quality, near-frontal; use only as a sanity check, not your headline result

Purpose	Dataset(s)	Notes
Face recognition — mixed quality	IJB-B, IJB-C	Where AdaFace's advantage is demonstrated
Face recognition — surveillance-native (recommended headline eval)	QMUL-SurvFace, QMUL-TinyFace	Real uncooperative surveillance imagery; this is what should anchor your "identity-aware evaluation" novelty claim
Object detection (traffic)	BDD100K, UA-DETRAC	Independent of the face branch
Multi-object tracking	MOT17, MOT20	Standard tracking benchmarks for ByteTrack/BoT-SORT validation
